Supercontinuum Generation in a Photonic Crystal Fibre

The generalised nonlinear Schrödinger equation with RK4IP and split step, the Dudley–Genty–Coen benchmark, and where the simple estimates fail

Kevin

Abstract

A femtosecond pulse launched into the anomalous-dispersion region of a photonic crystal fibre compresses, breaks into fundamental solitons, sheds dispersive waves and red-shifts by Raman scattering, spreading its spectrum over an octave. We solve the generalised nonlinear Schrödinger equation (GNLSE) with dispersion to tenth order, Kerr, self-steepening and the delayed Raman response, using two integrators (symmetric split-step Fourier and the fourth-order Runge–Kutta interaction-picture method, RK4IP), each with fixed or adaptive steps, plus a batched GPU implementation that reproduces the serial solver to 10^{-15}. The solver reproduces the NLS analytic limits to round-off (fundamental soliton shape error 10^{-10} over $20L_D$; $N = 2, 3$ recurrences at z_0 to 10^{-11}, 10^{-9}), converges at the theoretical orders $h^{4.02}$ and $h^{2.00}$, and reproduces the benchmark of Dudley *et al.* (fission at $0.95\,L_D/N$, $-40\,\mathrm{dB}$ span 491–1307 nm). Five further studies follow: the photon number is the true invariant of the full GNLSE (drift 3.5×10^{-7} while energy falls 8.9%); the Gordon self-frequency-shift formula underestimates short solitons by an excess $\propto T_0^{-1.95}$ (40% at 30 fs); compression and fission obey NLS scaling with soliton order; coherence collapses between $N \approx 9$ and 17; and an ablation study isolates the role of each higher-order term.

1 Introduction

Supercontinuum sources (white-light lasers spanning more than an octave) underpin optical coherence tomography, frequency metrology (carrier–envelope offset detection) and broadband spectroscopy. In photonic crystal fibres, the dominant mechanism for femtosecond pumping near the zero-dispersion wavelength is soliton fission followed by Raman self-frequency shift and dispersive-wave emission [1, 11]. All of this is captured by a single scalar equation, the GNLSE [2], but solving it accurately is demanding: the problem is stiff (tenth-order dispersion), multiscale (a $250\,\mu$m nonlinear length at peak compression inside a 15 cm fibre) and, for long pulses, noise-sensitive.

The aim of this project is a GNLSE solver whose accuracy is established step by step (analytic limits, invariants, convergence order, window and grid studies) before it is used to test the simple estimates that are commonly quoted: the Gordon formula, L_D/N fission, phase matching, "$N-1$ solitons", and the $N \lesssim 10$ coherence rule.

Contributions.

1. Two integrators (RK4IP, split step) with fixed and adaptive steps, a fixed Fourier/sign convention, and a batched GPU solver identical to the serial one.
2. Checks against the linear, SPM and soliton limits, convergence orders, and window/grid studies.
3. Reproduction of the Dudley–Genty–Coen benchmark with physical soliton classification.
4. Extensions: photon-number invariance, the Gordon-formula correction, soliton-order scaling, coherence and term ablation.

Table 1: Summary of the main results. Benchmark: 835 nm pump, $T_0 = 28.4$ fs (50 fs FWHM), $P_0 = 10$ kW, $\gamma = 0.11\,\mathrm{W^{-1}m^{-1}}$, 15 cm; $L_D = 6.82$ cm, $N = 8.66$, $L_D/N = 0.79$ cm.

Quantity	Computed	Analytic / literature
Raman time $T_R = f_R \int t\, h_R\, dt$	1.462 fs (quadrature)	closed form, Eq. (13)
L_D, N, zero-dispersion wavelength	6.82 cm, 8.66, 780 nm	Ref. [1]
Linear Gaussian broadening, $0 \dots 4L_D$	max rel. error 4×10^{-16}	$T_0\sqrt{1+(z/L_D)^2}$
SPM peaks at $\phi_{\max} = 1.5, 2.5, 3.5, 4.5\pi$	2, 3, 4, 5	M with $\phi_{\max} = (M - \frac{1}{2})\pi$
Fundamental soliton over $20L_D$	shape error 1.1×10^{-10}	shape invariant
$N = 2$ / $N = 3$ recurrence	$z_0(1 \pm 6 \times 10^{-12})$ / $z_0(1 \pm 4 \times 10^{-9})$	$z_0 = \frac{\pi}{2} L_D$
Energy (Kerr $+\beta_2..\beta_{10}$) through fission	RK4IP 4.5×10^{-10}, split step 2.9×10^{-13}	conserved
Self-frequency shift, $T_0 = 100$ fs	-0.0987 rad/ps/m	Gordon -0.0922 (7% lower)
Dispersive wave ($\beta_3 = 0.5$, $N = 1$)	peak $\Omega T_0 = 6.315$	phase matching 6.281
Benchmark fission point	$0.75\,\mathrm{cm} = 0.95\,L_D/N$	$\sim L_D/N$
Benchmark spectrum at -40 dB	491–1307 nm	Ref. [1]
Convergence order	RK4IP $h^{4.02}$, split step $h^{2.00}$	h^4, h^2
Photon number, full GNLSE	drift 3.5×10^{-7} (energy -8.9%)	conserved
RK4IP vs split step at equal cost	$6\times$ fewer evaluations at 10^{-6}, $20\times$ at 10^{-8}	—
Gordon excess vs T_0	$\propto T_0^{-1.95}$	$\propto T_0^{-2}$
NLS compression, $N = 16$	$z_{\mathrm{opt}}/z_0 = 0.0244$, $F_c = 65.9$	0.0243, 65.6
Coherence $\langle\lvert g_{12}\rvert\rangle$, 50/100/200 fs	0.97 / 0.17 / 0.07	—
Batched GPU vs serial	$\le 10^{-11}$ (typ. 10^{-15}), identical steps	identical

2 Theory and derivations

2.1 From the wave equation to the GNLSE

Write the field in the fibre mode as $E = \frac{1}{2}F(x,y)A(z,t)e^{i(\beta_0 z - \omega_0 t)} + \text{c.c.}$, with a slowly varying envelope A normalised so that $|A|^2$ is the power. Each frequency component $\omega = \omega_0 + \Omega$ propagates with the mode's wavenumber $\beta(\omega)$, which we expand about the carrier:

$$\beta(\omega_0 + \Omega) = \beta_0 + \beta_1\Omega + \sum_{k\ge 2} \frac{\beta_k}{k!}\Omega^k. \tag{1}$$

In the frame moving with the group velocity, $T = t - \beta_1 z$, the linear part of the propagation is $\tilde{A}(z,\Omega) \propto e^{i\beta(\Omega)z}$ with $\beta(\Omega) = \sum_{k\ge 2}\beta_k\Omega^k/k!$. The third-order nonlinear polarisation, with an instantaneous electronic (Kerr) part and a delayed vibrational (Raman) part, adds a term proportional to $A\int R(T')|A(T-T')|^2 dT'$. Keeping the first-order frequency dependence of the nonlinear coefficient gives the self-steepening factor $1 + (i/\omega_0)\partial_T$.

2.2 The generalised nonlinear Schrödinger equation

Units are ps, rad/ps, m and W. The envelope $A(z,T)$ obeys

$$\frac{\partial A}{\partial z} = -\frac{\alpha}{2}A + \sum_{k\ge 2}\frac{i^{k+1}\beta_k}{k!}\frac{\partial^k A}{\partial T^k} + i\gamma\Big(1 + \frac{i}{\omega_0}\frac{\partial}{\partial T}\Big)\Big[A\int R(T')\,|A(T-T')|^2\,dT'\Big], \tag{2}$$

with the Blow–Wood response [6]

$$R(T) = (1 - f_R)\,\delta(T) + f_R\,h_R(T), \qquad h_R(T) = \frac{\tau_1^2 + \tau_2^2}{\tau_1\tau_2^2}\,e^{-T/\tau_2}\sin(T/\tau_1)\,\Theta(T), \tag{3}$$

$f_R = 0.18$, $\tau_1 = 12.2$ fs and $\tau_2 = 32$ fs. h_R is normalised to unit area and is causal.

Fourier convention. We write $A(T) = \sum_\Omega \tilde{A}(\Omega) e^{-i\Omega T}$, so that $\partial_T \to -i\Omega$ and positive Ω is blue. In the frequency domain

$$\frac{d\tilde{A}}{dz} = \Big[i\beta(\Omega) - \frac{\alpha}{2}\Big]\tilde{A} + i\gamma\Big(1 + \frac{\Omega}{\omega_0}\Big)\,\mathcal{F}\Big\{A\big[(1-f_R)|A|^2 + f_R\,(h_R \circledast |A|^2)\big]\Big\}. \tag{4}$$

The sign of the Raman shift depends on this convention, and is fixed by a test.

2.3 Characteristic scales

Two lengths compare dispersion with nonlinearity:

$$L_D = \frac{T_0^2}{|\beta_2|}, \qquad L_{NL} = \frac{1}{\gamma P_0}, \qquad N^2 = \frac{L_D}{L_{NL}} = \frac{\gamma P_0 T_0^2}{|\beta_2|}. \tag{5}$$

The soliton period is $z_0 = \frac{\pi}{2} L_D$, and soliton fission occurs near $L_{\text{fiss}} \sim L_D/N$. For the benchmark ($\beta_2 = -1.183 \times 10^{-2}\,\text{ps}^2/\text{m}$): $L_D = 6.82\,\text{cm}$, $N = 8.66$ and $L_{\text{fiss}} \approx 0.79\,\text{cm}$.

2.4 Analytic limits

Dispersion only. With $\gamma = 0$ and only β_2, Eq. (4) gives $\tilde{A}(z,\Omega) = \tilde{A}(0,\Omega)\, e^{i\beta_2\Omega^2 z/2}$. For a chirp-free Gaussian $A_0 = e^{-T^2/2T_0^2}$ the transform is again Gaussian and the inverse transform gives

$$|A(z,T)|^2 \propto \exp\Big[-\frac{T^2}{T_0^2(1 + z^2/L_D^2)}\Big], \qquad T(z) = T_0\sqrt{1 + (z/L_D)^2}. \tag{6}$$

SPM only. With $\beta_k = 0$ and instantaneous Kerr, $\partial_z A = i\gamma|A|^2 A$. Then $\partial_z |A|^2 = 0$, so the intensity is unchanged and the solution is

$$A(z,T) = A(0,T)\, e^{i\gamma|A(0,T)|^2 z}. \tag{7}$$

The peak phase $\phi_{\max} = \gamma P_0 z$ produces a chirp, and each frequency in the broadened spectrum is generated at two times with a relative phase. The resulting interference gives M spectral peaks when $\phi_{\max} \approx (M - \frac{1}{2})\pi$.

SPM chirp. With the convention $A \propto e^{-i\Omega T}$, a time-dependent phase $\phi(T)$ shifts the local frequency by $\delta\Omega(T) = -\partial_T \phi$. For SPM,

$$\delta\Omega(T) = -\gamma z\, \partial_T |A(0,T)|^2. \tag{8}$$

The leading edge of the pulse, where the intensity rises, is shifted to the red and the trailing edge to the blue; the maximum shift is of order $\phi_{\max}/T_0$.

Fundamental soliton. For $\beta_2 < 0$, insert $A = \sqrt{P_0}\,\text{sech}(T/T_0)\, e^{iz/2L_D}$ into the NLS, $\partial_z A = -\frac{i\beta_2}{2}\partial_T^2 A + i\gamma|A|^2 A$. Using

$$\partial_T^2\,\text{sech}(T/T_0) = \frac{1}{T_0^2}\big(\text{sech} - 2\,\text{sech}^3\big), \tag{9}$$

the right-hand side is

$$\frac{i|\beta_2|}{2T_0^2}\,\text{sech} \;+\; i\Big(\gamma P_0 - \frac{|\beta_2|}{T_0^2}\Big)\text{sech}^3, \tag{10}$$

times $\sqrt{P_0}$. The sech^3 term vanishes iff $\gamma P_0 T_0^2 = |\beta_2|$, i.e. $N = 1$, and the remaining term equals the left-hand side $(i/2L_D)A$. The pulse propagates without changing shape.

Higher-order solitons. The NLS is integrable. The input $N\sqrt{P_0}\,\mathrm{sech}(T/T_0)$ with integer N is a bound state of N solitons with equal velocities but different nonlinear phases, which beat with a common period $z_0 = \frac{\pi}{2}L_D$. The intensity profile recurs exactly at z_0.

2.5 Invariants: energy and photon number

Take the time derivative of $\sum_\Omega |\tilde{A}|^2/(\omega_0+\Omega)$ with Eq. (4) and $\alpha = 0$.

- The dispersion term contributes $2\,\mathrm{Re}[i\beta(\Omega)]|\tilde{A}|^2/(\omega_0+\Omega) = 0$.
- In the nonlinear term the factor $(1+\Omega/\omega_0) = (\omega_0+\Omega)/\omega_0$ cancels the weight. With $V = R \circledast |A|^2$ and Parseval's theorem, what remains is

$$\frac{2\gamma}{\omega_0}\,\mathrm{Re}\Big[i\sum_\Omega \tilde{A}^*\mathcal{F}\{AV\}\Big] = \frac{2\gamma}{\omega_0}\,\mathrm{Re}\Big[i\int |A|^2 V\,dT\Big],$$

  which vanishes because V is real.

Hence the full GNLSE, including self-steepening and Raman scattering, conserves the photon number

$$\mathcal{N}_{\mathrm{ph}} \propto \sum_\Omega \frac{|\tilde{A}(\Omega)|^2}{\omega_0+\Omega}. \tag{11}$$

The energy $E = \int |A|^2 dT$ is conserved by dispersion, Kerr and self-steepening, but not by Raman scattering: each Stokes-scattered photon leaves part of its energy in a phonon. Energy therefore tests only part of the solver; photon number tests all of it.

2.6 Raman time

The first moment of the response defines the Raman time $T_R = f_R\int_0^\infty t\,h_R(t)\,dt$. Writing $\sin(t/\tau_1) = \mathrm{Im}\,e^{it/\tau_1}$ and $s = 1/\tau_2 - i/\tau_1$,

$$\int_0^\infty t\,e^{-t/\tau_2}\sin(t/\tau_1)\,dt = \mathrm{Im}\,\frac{1}{s^2} = \frac{2\tau_1^3\tau_2^3}{(\tau_1^2+\tau_2^2)^2}. \tag{12}$$

Multiplying by the prefactor $(\tau_1^2+\tau_2^2)/(\tau_1\tau_2^2)$ of h_R,

$$T_R = f_R\,\frac{2\tau_1^2\tau_2}{\tau_1^2+\tau_2^2} = 1.462\ \mathrm{fs}. \tag{13}$$

2.7 Raman gain spectrum

The response in the frequency domain, $\tilde{h}_R(\Omega) = \int_0^\infty h_R(t)\,e^{i\Omega t}\,dt$, follows from the same integral as above with $s = 1/\tau_2 - i\Omega$:

$$\tilde{h}_R(\Omega) = \frac{\tau_1^2+\tau_2^2}{\tau_1^2\,(1-i\Omega\tau_2)^2+\tau_2^2}, \qquad \tilde{h}_R(0) = 1. \tag{14}$$

Its imaginary part is odd in Ω and peaks near $|\Omega| \approx 1/\tau_1 = 82\,\mathrm{rad/ps}$ (13 THz). It is the Raman gain spectrum: light on the Stokes (red) side of a strong field is amplified and light on the anti-Stokes side is attenuated. The Raman time is the slope of $\tilde{h}_R$ at the origin,

$$T_R = f_R\int_0^\infty t\,h_R\,dt = -i\,f_R\,\frac{d\tilde{h}_R}{d\Omega}\Big|_{\Omega=0}, \tag{15}$$

and differentiating the closed form reproduces Eq. (13). A soliton whose spectrum is narrow compared with 13 THz sees only this linear slope, which is the approximation behind the Gordon formula below.

2.8 Soliton self-frequency shift

First-order Raman term. For pulses much longer than the response time, $h_R \circledast |A|^2 \approx |A|^2 - \bar{\tau}\,\partial_T|A|^2$ with $\bar{\tau} = \int t\,h_R\,dt$, so the Raman part of Eq. (2) adds

$$\partial_z A\big|_{\mathrm{R}} = -i\gamma T_R\,A\,\partial_T|A|^2. \tag{16}$$

Mean frequency. With our convention the mean frequency is $\bar{\Omega} = E^{-1}\int A^*\,(i\partial_T A)\,dT$. The NLS terms conserve it. Differentiating and integrating by parts, the Raman term gives

$$\frac{d\bar{\Omega}}{dz} = -\frac{2}{E}\,\mathrm{Im}\int \big(\partial_z A\big)^*\partial_T A\,dT = -\frac{\gamma T_R}{E}\int \big(\partial_T|A|^2\big)^2 dT, \tag{17}$$

where we used $\mathrm{Re}(A^*\partial_T A) = \frac{1}{2}\partial_T|A|^2$. The integrand is positive, so the shift is always to the red, a consequence of causality.

Fundamental soliton. For $|A|^2 = P_0\,\mathrm{sech}^2 x$ with $x = T/T_0$, $E = 2P_0T_0$ and

$$\int \big(\partial_T|A|^2\big)^2 dT = \frac{4P_0^2}{T_0}\int \mathrm{sech}^4 x\,\tanh^2 x\,dx = \frac{4P_0^2}{T_0}\cdot\frac{4}{15}. \tag{18}$$

Using $\gamma P_0 = |\beta_2|/T_0^2$ gives Gordon's formula [7]

$$\frac{d\Omega_s}{dz} = -\frac{8\,T_R\,|\beta_2|}{15\,T_0^4}, \tag{19}$$

which for $T_0 = 100\,\mathrm{fs}$ is $-0.0922\,\mathrm{rad/ps/m}$. The derivation linearises the Raman gain around $\Omega = 0$. A short soliton has a broad spectrum that already samples the curved, rising part of the gain, so the first correction should scale as $(\Omega_{\mathrm{soliton}}/\Omega_R)^2 \propto T_0^{-2}$.

2.9 Soliton fission and dispersive waves

An N-th order soliton compresses until it reaches its narrowest point, near $z \approx L_D/N$. Non-integrable perturbations (β_3, Raman, self-steepening) then split it into fundamental solitons, which in the inverse-scattering picture have peak powers

$$P_k \approx P_0\,\frac{(2N-2k+1)^2}{N^2}, \qquad k = 1, 2, \ldots \tag{20}$$

plus radiation. The most powerful soliton is the narrowest, and therefore Raman-shifts the most.

Phase matching. In the moving frame a soliton of centre frequency Ω_s and peak power P_s has a wavenumber linear in frequency (it is non-dispersive), $\beta(\Omega_s) + \beta_1(\Omega_s)(\Omega - \Omega_s) + \gamma P_s/2$, where $\gamma P_s/2$ is its nonlinear phase per unit length. A linear wave has wavenumber $\beta(\Omega)$. Energy is transferred resonantly where the two agree [8]:

$$\beta(\Omega) - \beta(\Omega_s) - \beta_1(\Omega_s)(\Omega - \Omega_s) = \frac{\gamma P_s}{2}. \tag{21}$$

With only β_2 and β_3, $\Omega_s = 0$ and small P_s, this gives $\Omega \approx -3\beta_2/\beta_3$, on the normal-dispersion side. When radiation is emitted, momentum conservation pushes the soliton the other way (recoil); the soliton's *actual* centre frequency and peak power must then be used in Eq. (21).

2.10 NLS compression scaling

For the pure NLS the point of maximum compression and the compression factor follow the empirical laws [1]

$$\frac{z_{\mathrm{opt}}}{z_0} \approx \frac{0.32}{N} + \frac{1.1}{N^2}, \qquad F_c \approx 4.1N. \tag{22}$$

2.11 Modulation instability and coherence

A continuous wave of power P_0 in anomalous dispersion is unstable to small modulations. Linearising the NLS about it, a perturbation at frequency offset Ω grows with power gain

$$g(\Omega) = |\beta_2 \Omega| \sqrt{\Omega_c^2 - \Omega^2}, \qquad \Omega_c^2 = \frac{4\gamma P_0}{|\beta_2|}, \tag{23}$$

with maximum $g_{\max} = 2\gamma P_0$. Over the fission length L_D/N, noise is amplified in amplitude by

$$\exp\left(\frac{g_{\max}}{2}\,\frac{L_D}{N}\right) = e^N.$$

For small N deterministic fission happens first and the output is reproducible from shot to shot; for large N, noise-seeded modulation instability breaks the pulse before fission and the spectral phase becomes random. The shot-to-shot stability is measured by the first-order coherence over an ensemble with one photon per mode of random phase added to the input [10]:

$$|g_{12}(\lambda)| = \frac{\left|\langle E_i^*(\lambda) E_j(\lambda)\rangle_{i\neq j}\right|}{\langle |E(\lambda)|^2\rangle}, \tag{24}$$

reported as a spectrum-weighted mean over the $-40\,\mathrm{dB}$ bandwidth.

3 Numerical methods

3.1 Discretisation

The envelope is sampled on a periodic time grid of n points over a window T_w, so the frequency grid has spacing $2\pi/T_w$ and spans $2\pi n/T_w$. Derivatives in T become multiplications in frequency, and the problem is written as

$$\frac{d\tilde{A}}{dz} = \hat{L}\tilde{A} + \hat{N}(\tilde{A}), \qquad \hat{L} = i\beta(\Omega) - \frac{\alpha}{2}, \tag{25}$$

with $\hat{L}$ diagonal in frequency. The linear part is stiff: $\beta(\Omega)$ contains terms up to Ω^{10}, which are large at the window edges. The nonlinear part is evaluated pseudo-spectrally: transform to time, form $A \cdot V$, transform back and multiply by $i\gamma(1 + \Omega/\omega_0)$.

Raman convolution. h_R is sampled from $t = 0$ and normalised so that $\sum h_R\, dt = 1$. The combined response $(1 - f_R)I + f_R\, h_R \circledast I$ of the intensity $I = |A|^2$ is applied as a single multiplication in the frequency domain with real FFTs.

3.2 RK4IP

In the interaction picture $A_I = e^{-\hat{L}(z-z_0)}A$, with z_0 at the centre of the step, the linear part is integrated exactly and only the nonlinear part is left for a Runge–Kutta method [3]:

$$\begin{aligned}
A_I &= e^{h\hat{L}/2}A, & k_1 &= e^{h\hat{L}/2}\big[h\hat{N}(A)\big],\\
k_2 &= h\hat{N}\big(A_I + \tfrac{1}{2}k_1\big), & k_3 &= h\hat{N}\big(A_I + \tfrac{1}{2}k_2\big),\\
k_4 &= h\hat{N}\big(e^{h\hat{L}/2}(A_I + k_3)\big),\\
A(z+h) &= e^{h\hat{L}/2}\Big[A_I + \tfrac{1}{6}k_1 + \tfrac{1}{3}k_2 + \tfrac{1}{3}k_3\Big] + \tfrac{1}{6}k_4.
\end{aligned} \tag{26}$$

Each step costs four nonlinear evaluations and the global error is $O(h^4)$. Because the linear part is exact, the step is not limited by the stiffness of $\hat{L}$.

3.3 Split-step Fourier

Strang splitting alternates the two operators symmetrically,

$$A(z+h) \approx e^{h\hat{L}/2}\, e^{h\hat{N}}\, e^{h\hat{L}/2} A(z), \tag{27}$$

with global error $O(h^2)$ from the commutator $[\hat{L}, \hat{N}]$. For pure Kerr the nonlinear sub-step is exact, $A\, e^{i\gamma|A|^2 h}$, and the scheme is unitary. With self-steepening and Raman the nonlinear sub-step is integrated by classical RK4 (four evaluations), so both methods can be compared at equal cost.

3.4 Adaptive stepping

RK4IP. The embedded RK4(3)IP scheme of Balac and Mahé [4] computes a third-order solution u_3 alongside the fourth-order u_4, reusing the last evaluation of a step as the first of the next (first-same-as-last), so an accepted step still costs four evaluations. The local error is estimated as

$$\varepsilon = \frac{\|u_4 - u_3\|}{\|u_4\|} = \frac{h}{10}\, \frac{\|k_4 - k_5\|}{\|u_4\|}, \tag{28}$$

and the next step is $h \to h \cdot 0.9\,(\mathrm{tol}/\varepsilon)^{1/4}$, clamped to $[0.2, 2]$. A step shortened to land on a save point does not shrink the next one.

Split step. Step doubling: one step of size h and two of $h/2$ are compared, which gives a local error estimate [5]; the two results are combined by Richardson extrapolation. A step is rejected if its error exceeds $2\,\mathrm{tol}$, and the step size changes by factors of $2^{1/3}$.

3.5 Batched integration

Many independent propagations (noise seeds, parameter scans) are advanced together as one array of shape (runs, n). Each row keeps its own grid, fibre, pulse, tolerance, adaptive step and save points, and finished rows leave the batch. The arithmetic per row is identical to the serial solver, so in double precision the results agree to $\sim 10^{-15}$ with identical step counts. On a laptop GPU (RTX 3050) a double-precision FFT costs about the same for 1 field as for 16, so the speed-up comes from batching: 16 noise seeds at $n = 2^{14}$ run about $6\times$ faster than serially on the CPU.

3.6 Input noise

For the coherence study each run adds one photon per spectral mode with a random phase to the input field [10]. A mode of the discrete spectrum has energy $T_w|\tilde{A}_k|^2$, so one photon of energy $\hbar(\omega_0 + \Omega_k)$ corresponds to

$$|\tilde{A}_k^{\mathrm{noise}}| = \sqrt{\frac{\hbar(\omega_0 + \Omega_k)}{T_w}}, \qquad \arg \tilde{A}_k^{\mathrm{noise}} \text{ uniform on } [0, 2\pi). \tag{29}$$

Sixteen independent seeds are propagated per pulse duration, and $|g_{12}|$ is evaluated from all pairs.

3.7 Spectrogram

The time–frequency picture of the output is the gated spectrogram

$$S(\tau, \Omega) = \left| \int A(T)\, g(T - \tau)\, e^{i\Omega T}\, dT \right|^2, \tag{30}$$

with a Gaussian gate g of 50 fs FWHM, evaluated on 300 delays between -0.5 and 5 ps by one FFT per delay.

3.8 Aliasing of the nonlinear term

The Kerr term $|A|^2A$ contains frequencies up to three times the spectral width of A. On a finite frequency grid, the part that exceeds the grid is folded back (aliased) onto the other end of the spectrum. The grid must therefore leave an empty margin around the spectrum. This is monitored by the fraction of energy within 5% of each end of the frequency window, which must stay negligible; the grid study in Section 5.4 shows that 2^{12} points over 12.5 ps fail this test and 2^{13} pass it.

3.9 Diagnostics

- *Pulse width:* RMS width of $|A|^2$. *Mean frequency:* first moment of $|\tilde{A}|^2$, compared with Eq. (19) through its slope in z.
- *Fission point:* the z of maximum peak power.
- *Soliton classifier:* a temporal peak counts as a fundamental soliton if its gated spectrum lies in the anomalous region, its local soliton number $\sqrt{\gamma P T_0^2/|\beta_2(\Omega_c)|}$ is between 0.6 and 1.6, its width is at least four grid points, and it is not a shoulder within $4T_0$ of a stronger soliton.
- *Phase matching:* Eq. (21) solved with the full Taylor series and the measured Ω_s, P_s.
- *Window checks:* the fraction of energy within 5% of each end of the time and frequency windows, recorded for every run.
- *Convergence:* the error at fixed z against a reference with a much smaller step; the order is the slope of log(error) against $\log h$ in the range where the error is between 10^{-12} and 10^{-4}.

4 Verification

The checks are ordered from limits with exact solutions to the full equation, so that each new term is tested only after the parts it builds on.

1. Conventions and scales. With $\partial_T \to -i\Omega$, positive Ω must be blue; this fixes the sign of every frequency shift. The quadrature value of T_R must match Eq. (13), and L_D, N and the zero-dispersion wavelength must match the published benchmark.

2. Exact limits. Each limit isolates one operator:

- dispersion only: the RMS width must follow $T_0\sqrt{1+(z/L_D)^2}$, which tests $\hat{L}$ and the FFT grid;
- SPM only: the number of spectral peaks must be M at $\phi_{\max} = (M - \frac{1}{2})\pi$, which tests $\hat{N}$;
- fundamental soliton: the shape must be unchanged over $20L_D$, which tests the balance of the two;
- $N = 2, 3$: the input profile must recur at z_0, which tests the phase relations between the bound solitons and is sensitive to small errors.

3. Conservation laws. With dispersion and Kerr only, the energy must be conserved through fission: exactly (up to rounding) for the unitary split step, and to the truncation error for RK4IP. With the full GNLSE the photon number must be conserved while the energy changes. Since the conservation follows from the structure of Eq. (4) rather than from any particular solution, it tests the self-steepening factor and the Raman convolution together.

4. Perturbative predictions. The self-frequency shift of a 100 fs soliton must agree with Eq. (19) (where the formula is expected to hold), and the dispersive-wave peak must agree with Eq. (21) within half its width. Both depend on the sign convention of point 1.

5. Benchmark and numerics. The fission point and the −40 dB span must agree with Dudley *et al.* The two integrators must agree with each other, must converge as h^4 and h^2, and the batched solver must reproduce the serial one. Window and grid convergence are measured explicitly (Section 5.4).

5 Results

5.1 Analytic limits

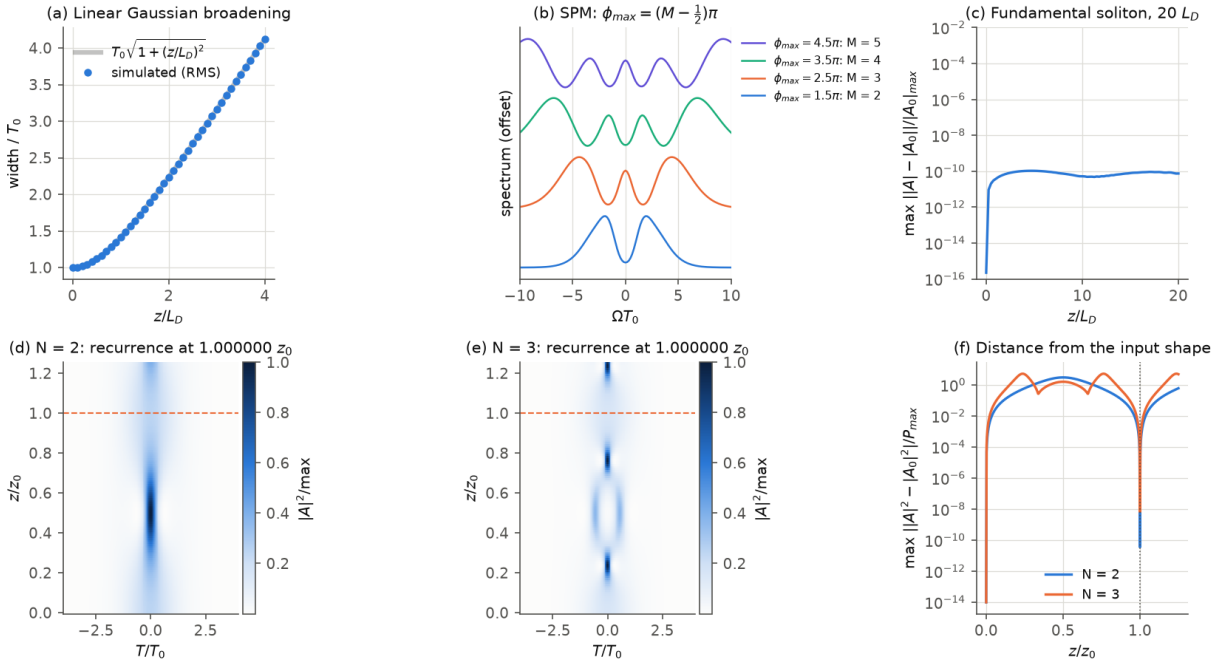


Figure 1: (a) RMS width under dispersion vs $T_0\sqrt{1+(z/L_D)^2}$. (b) SPM spectra at $\phi_{\max} = 1.5 \ldots 4.5\pi$. (c) Fundamental-soliton shape error over $20L_D$. (d, e) $N = 2$, $N = 3$ solitons over $1.25z_0$. (f) Distance from the input shape; the minimum at z_0 is the recurrence.

The linear and soliton limits are reproduced to round-off and to the integrator truncation error (fixed RK4IP, 4000 steps over $20L_D$, 8000 over $1.25z_0$).

5.2 Conservation, Raman shift and dispersive wave

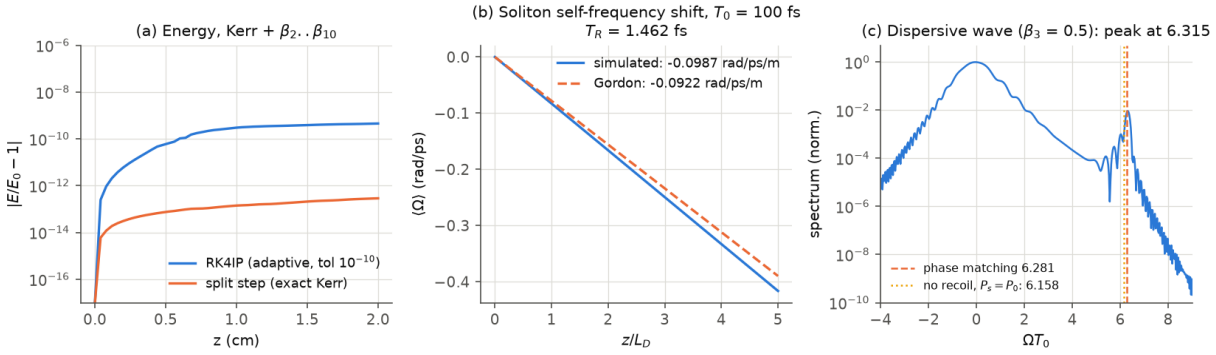


Figure 2: (a) Energy drift, Kerr $+\beta_2 \ldots \beta_{10}$, benchmark pulse over 2 cm: adaptive RK4IP (tol 10^{-10}) and exact-Kerr split step. (b) Mean frequency of a $T_0 = 100$ fs soliton vs the Gordon rate. (c) $N = 1$ soliton with $\beta_3 = 0.5$ after $10L_D$, with phase-matched frequency with and without recoil.

The split step conserves energy to 2.9×10^{-13}, i.e. to rounding, because it is unitary; RK4IP's drift (4.5×10^{-10}) is its truncation error at tol $= 10^{-10}$. Phase matching with the input values predicts $\Omega T_0 = 6.158$, outside the radiation peak; with the soliton's recoiled frequency and actual peak power it predicts 6.281 against the measured 6.315 (FWHM 0.176).

5.3 The Dudley–Genty–Coen benchmark

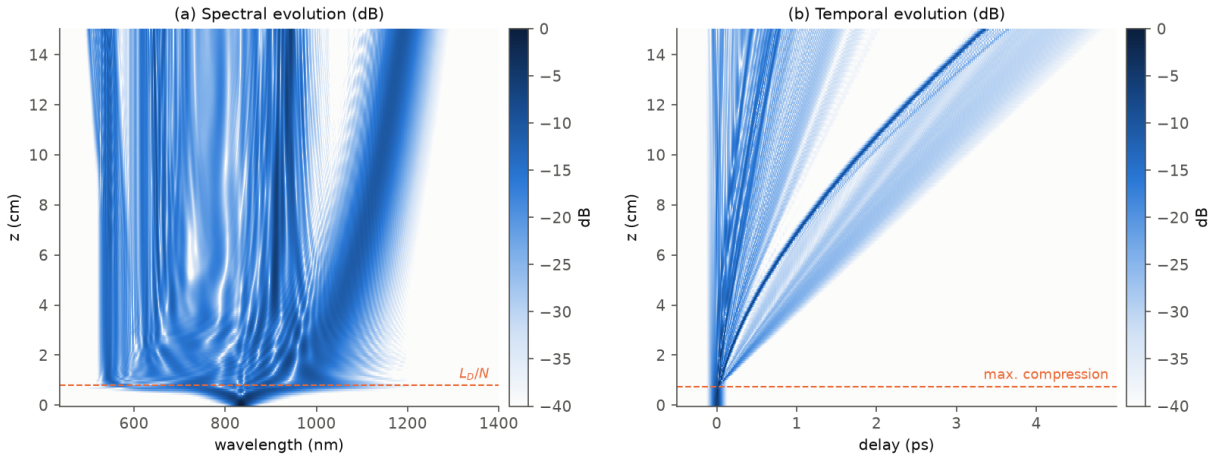


Figure 3: Benchmark over 15 cm on 2^{13} points over 12.5 ps (adaptive RK4IP, tol 10^{-8}: 10 369 steps). (a) Spectrum (dB) vs wavelength and z; dashed: L_D/N. (b) Temporal intensity; dashed: maximum compression.

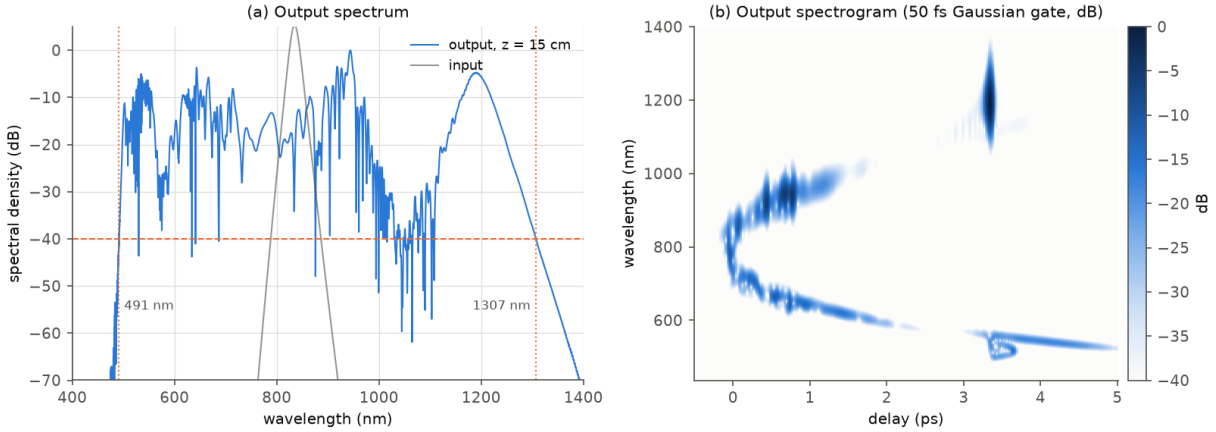


Figure 4: (a) Output and input spectra; dotted: −40 dB edges. (b) Output spectrogram (50 fs Gaussian gate).

Peak power is maximal at $z = 0.75$ cm (37 kW, $3.7P_0$), i.e. $0.95\,L_D/N$. The output spans 491–1307 nm at −40 dB, as in the paper. The leading Raman soliton peaks near 1190 nm and arrives 3.3 ps late; dispersive waves form the 500–600 nm band, and the spectrogram shows blue radiation trapped behind the red soliton [9].

Counting solitons. The physical classifier finds 5 fundamental solitons: the leading one (4.8 kW, $T_0 = 16$ fs, 1148 nm, at 3.35 ps) and four below 1 ps. The estimate "$N - 1 \approx 8$" also counts weak overlapping solitons that cannot be separated cleanly; a plain threshold count gives anything from 3 to 12.

5.4 Convergence

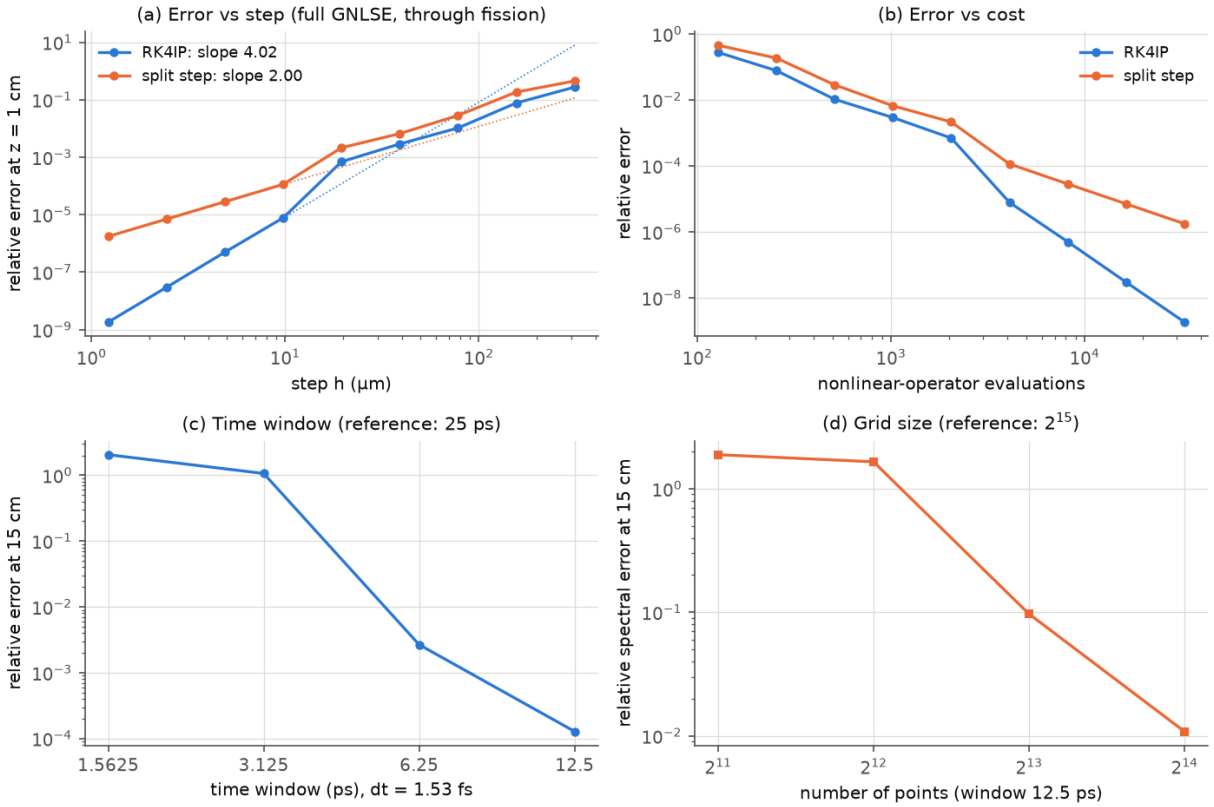


Figure 5: (a) Error at $z = 1\,\mathrm{cm}$ vs fixed step, full GNLSE; dotted: h^4, h^2. (b) Same error vs number of nonlinear evaluations. (c) Error at 15 cm vs time window. (d) Spectral error at 15 cm vs number of points.

For $h \lesssim 10\,\mu\mathrm{m}$ RK4IP converges as $h^{4.02}$ and split step as $h^{2.00}$; above $20\,\mu\mathrm{m}$ both are pre-asymptotic because the step does not resolve the $\sim 250\,\mu\mathrm{m}$ nonlinear length at peak compression. At equal error RK4IP needs $\sim 7\,000$ evaluations for 10^{-6} against $\sim 43\,000$ (6×), and $\sim 21\,000$ against $\sim 430\,000$ for 10^{-8} (20×).

time window (ps)	1.5625	3.125	6.25	12.5
error vs 25 ps	2.1	1.1	2.7×10^{-3}	1.3×10^{-4}
points (12.5 ps)	2^{11}	2^{12}	2^{13}	2^{14}
spectral error vs 2^{15}	1.9	1.7	9.7×10^{-2}	1.1×10^{-2}
energy at spectral edges	8×10^{-2}	4×10^{-2}	3×10^{-9}	6×10^{-11}

The Raman soliton walks 3.3 ps, so windows ≤ 3 ps fail completely; grids of $\leq 2^{12}$ points alias. At 2^{13} the dB spectrum and every tabulated number are converged; 2^{14} is needed for the complex field to 1%.

5.5 Photon number as the invariant

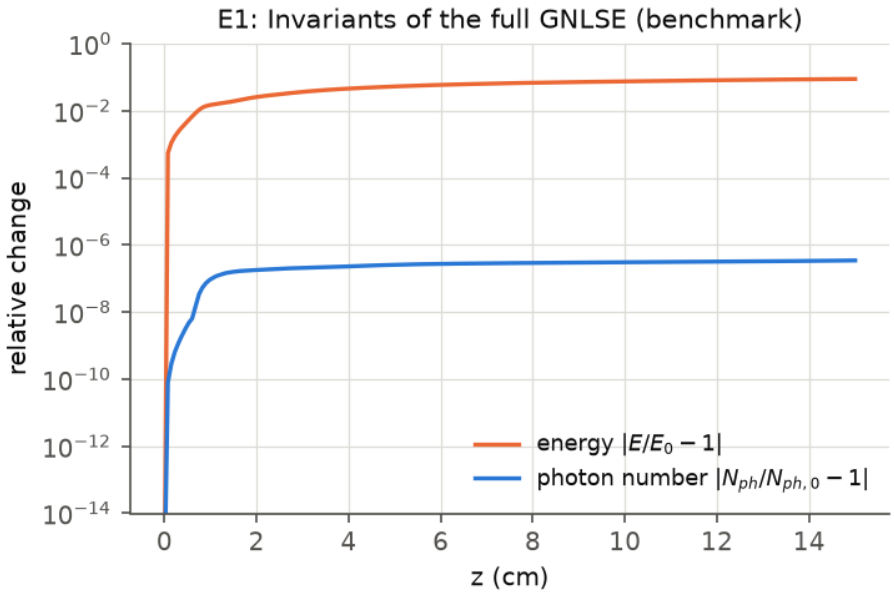


Figure 6: Relative change of energy and photon number along the benchmark (full GNLSE, tol 10^{-8}).

The photon number drifts by 3.5×10^{-7} over 15 cm while the energy drops by 8.9%.

5.6 Gordon formula and soliton-order scaling

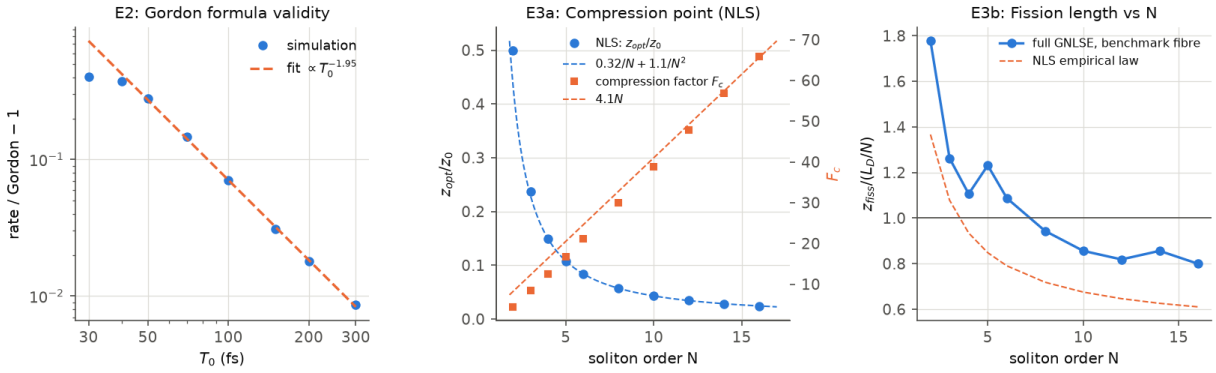


Figure 7: (a) Simulated shift rate over Gordon rate minus one, $T_0 = 30$–300 fs, with power-law fit for $T_0 \geq 70$ fs. (b) Pure NLS: maximum-compression point and factor vs N. (c) Full GNLSE: fission distance over L_D/N vs N.

The excess of the simulated shift over the Gordon formula falls as $T_0^{-1.95}$, close to the expected T_0^{-2} curvature correction: 40% at 30 fs, 28% at 50 fs, 7% at 100 fs, 0.9% at 300 fs. Compression follows $z_{\mathrm{opt}}/z_0 = 0.32/N + 1.1/N^2$ and $F_c \approx 4.1N$ ($N = 16$: 0.0244 and 65.9 vs 0.0243 and 65.6). The fission distance stays within a factor 1.5 of L_D/N for all $N \geq 3$, slightly beyond the pure-NLS law because higher-order dispersion and Raman delay fission.

5.7 Coherence versus pulse duration

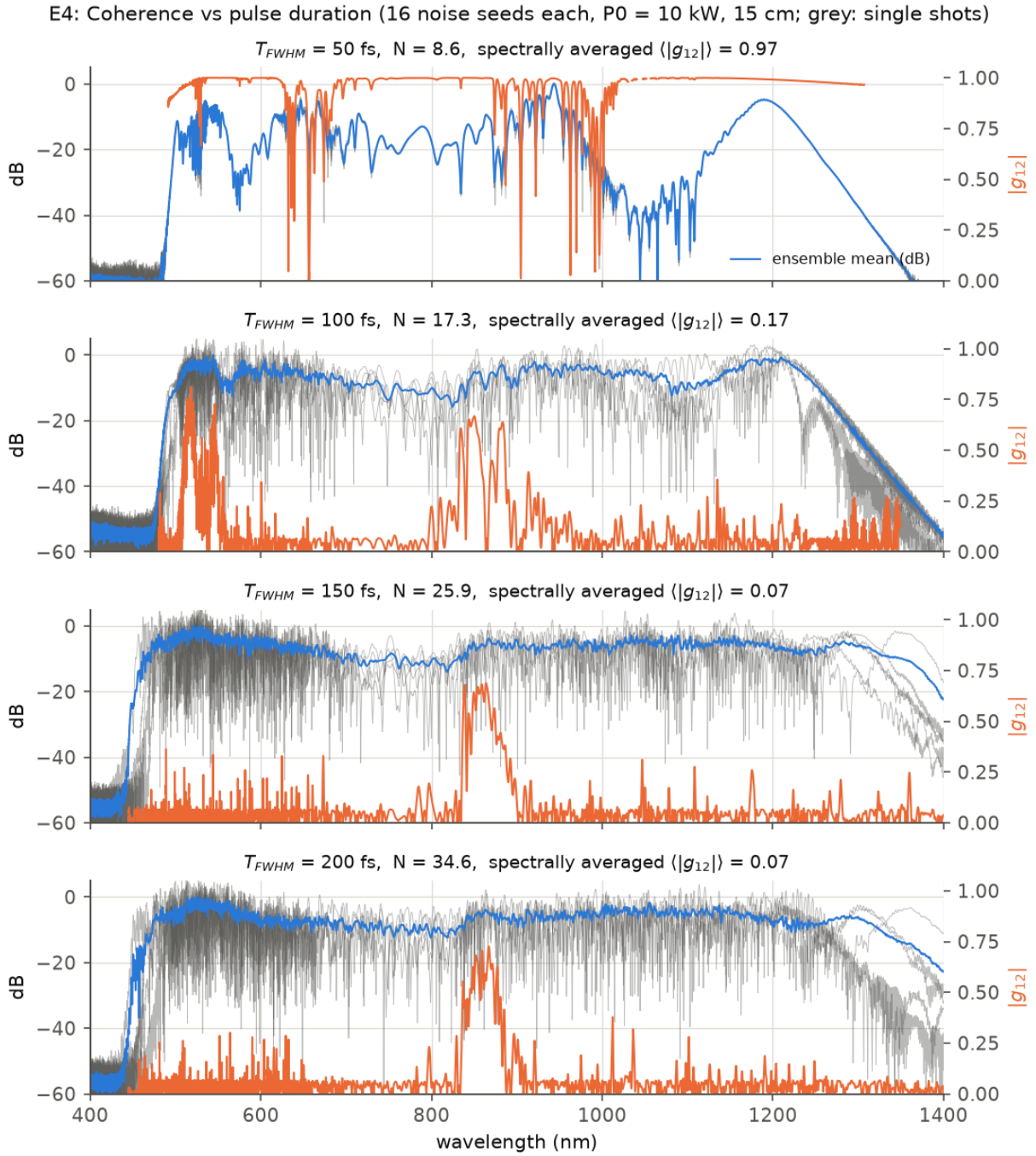


Figure 8: Ensemble-mean spectrum, five single shots (grey) and $|g_{12}(\lambda)|$ for 16 noise seeds per duration, $P_0 = 10\,\mathrm{kW}$, 15 cm.

T_{FWHM}	N	$\langle\lvert g_{12}\rvert\rangle$	energy at time-window edges
50 fs	8.6	0.97	3×10^{-6}
100 fs	17.3	0.17	1×10^{-6}
150 fs	25.9	0.07	6×10^{-6}
200 fs	34.6	0.07	5×10^{-6}

The transition between $N \approx 9$ and 17 is sharp, as expected from the e^N amplification of noise before fission. Longer pulses need a 25 ps window; on 12.5 ps, 16–19% of their energy wrapped around.

5.8 Contribution of each term

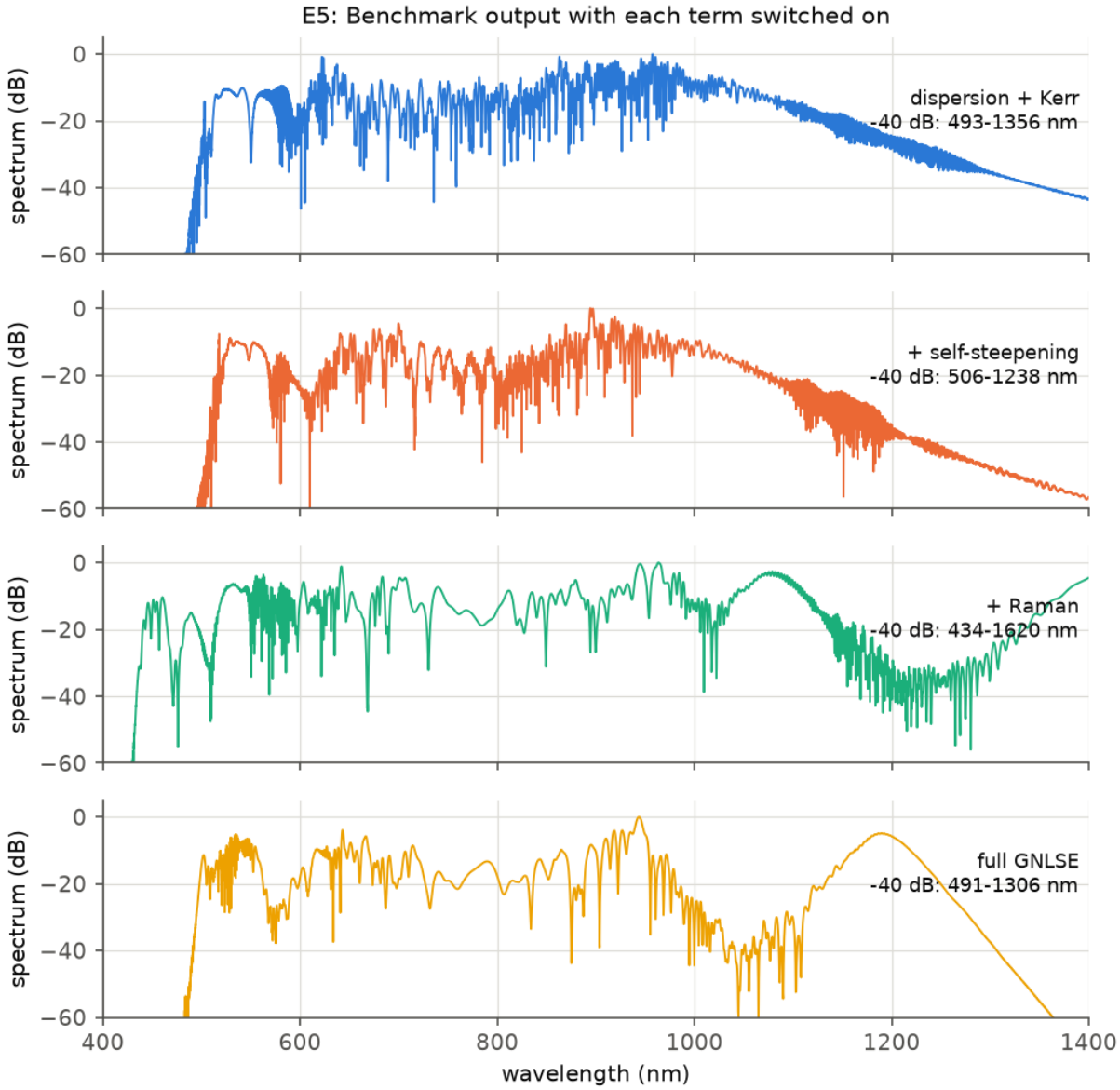


Figure 9: Benchmark output (15 cm, 2^{14} points) with higher-order terms switched on in turn.

model	−40 dB span	adaptive steps
dispersion + Kerr	493–1356 nm	13 215
+ self-steepening	506–1238 nm	7 905
+ Raman (no self-steepening)	434–1620 nm	11 146
full GNLSE	491–1306 nm	5 791

Without Raman there is no distinct red soliton; self-steepening moves energy to the blue during compression and lowers soliton peak powers; Raman without self-steepening overestimates broadening. Both are needed for the 490–1310 nm span.

6 Discussion

Corrections to common estimates. (1) $T_R = 1.462\,\mathrm{fs}$, not the often-quoted 1.5 fs. (2) The Gordon formula is not accurate to 20% at benchmark durations; this requires $T_0 \gtrsim 60\,\mathrm{fs}$. (3) Energy conservation to 10^{-8} is a property of the unitary split step; for RK4IP it is an accuracy check. (4) Phase matching needs the soliton recoil. (5) "$N-1$ solitons" is not a well-defined count without a physical criterion.

Why RK4IP wins. It integrates the stiff linear part exactly and approximates only the nonlinear part, to fourth rather than second order; the high-order dispersion at the window edges costs it no step-size penalty. Cost comparisons are made at equal numbers of nonlinear evaluations, which is independent of the machine.

Limitations. Constant γ and effective area (no $\gamma(\omega)$); no spontaneous Raman noise in the coherence study (it would lower coherence further); double precision is slow on consumer GPUs, so single runs gain nothing from the GPU, only batches do.

Outlook. Gas-filled hollow-core fibres (long-lived molecular Raman coherence, photoionisation), Kerr microresonator combs (the Lugiato–Lefever equation is this GNLSE with driving, detuning and round-trip loss), and realistic noise sources are natural next steps.

7 Conclusion

A GNLSE solver with two integrators, adaptive stepping and a batched GPU path was checked against every NLS analytic limit, invariant and convergence order before reproducing the Dudley–Genty–Coen benchmark. Using it, we showed that photon number, not energy, is the invariant that tests the full equation, quantified the T_0^{-2} breakdown of the Gordon formula, confirmed the NLS compression and fission scaling, located the coherence collapse between $N \approx 9$ and 17, and separated the roles of self-steepening and Raman scattering in the final spectrum.

References

[1] J. M. Dudley, G. Genty, S. Coen, *Supercontinuum generation in photonic crystal fiber*, Rev. Mod. Phys. **78**, 1135 (2006).

[2] G. P. Agrawal, *Nonlinear Fiber Optics*, 5th ed., Academic Press (2013).

[3] J. Hult, *A fourth-order Runge–Kutta in the interaction picture method for simulating supercontinuum generation in optical fibers*, J. Lightwave Technol. **25**, 3770 (2007).

[4] S. Balac, F. Mahé, *Embedded Runge–Kutta scheme for step-size control in the interaction picture method*, Comput. Phys. Commun. **184**, 1211 (2013).

[5] O. V. Sinkin, R. Holzlöhner, J. Zweck, C. R. Menyuk, *Optimization of the split-step Fourier method in modeling optical-fiber communications systems*, J. Lightwave Technol. **21**, 61 (2003).

[6] K. J. Blow, D. Wood, *Theoretical description of transient stimulated Raman scattering in optical fibers*, IEEE J. Quantum Electron. **25**, 2665 (1989).

[7] J. P. Gordon, *Theory of the soliton self-frequency shift*, Opt. Lett. **11**, 662 (1986).

[8] N. Akhmediev, M. Karlsson, *Cherenkov radiation emitted by solitons in optical fibers*, Phys. Rev. A **51**, 2602 (1995).

[9] A. V. Gorbach, D. V. Skryabin, *Light trapping in gravity-like potentials and expansion of supercontinuum spectra in photonic-crystal fibres*, Nat. Photonics **1**, 653 (2007).

[10] J. M. Dudley, S. Coen, *Coherence properties of supercontinuum spectra generated in photonic crystal and tapered optical fibers*, Opt. Lett. **27**, 1180 (2002).

[11] D. V. Skryabin, A. V. Gorbach, *Colloquium: Looking at a soliton through the prism of optical supercontinuum*, Rev. Mod. Phys. **82**, 1287 (2010).